



Government of **Western Australia**
Child and Adolescent Health Service



Deploying LLM web applications on Open OnDemand: experiences at Perth Children's Hospital

Dr Harry Smallbone

Dr Thomas Drake-Brockman

Child and Adolescent Health Service / University of Western Australia

Compassion

Excellence

Collaboration

Accountability

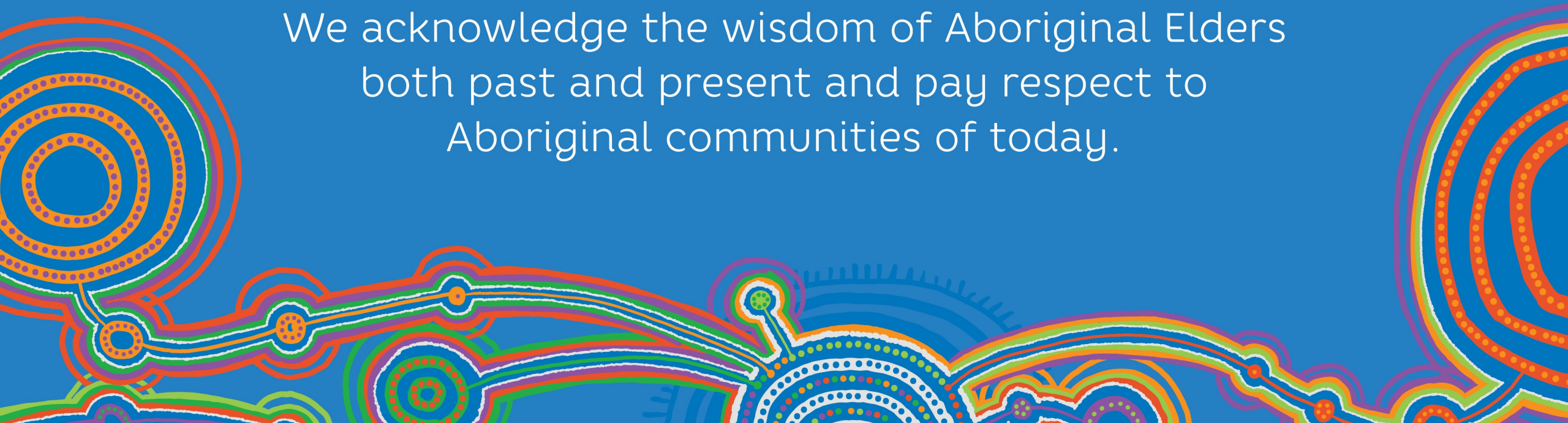
Equity

Respect

Acknowledgement of Country

The Child and Adolescent Health Service acknowledge
Aboriginal people of the many traditional lands and
language groups of Western Australia.

We acknowledge the wisdom of Aboriginal Elders
both past and present and pay respect to
Aboriginal communities of today.



Acknowledgements

- Thanks to GOOD for helping to fund my travel costs



Agenda

- Background
- Deploying Open OnDemand - design choices
- Standardising LLM applications
- Future work



Background

- Thomas and I are practicing medical doctors from Perth, Australia's most isolated capital city



pawsey



Background

Experience



Goose farmer

Self-employed

Jul 2023 - Present · 10 mos

On-site



Microsoft

22 yrs 4 mos



Principal Performance Architect

Full-time

Jul 2022 - Jul 2023 · 1 yr 1 mo

Chehalis, Washington, United States



Principal Software Development Engineer

Apr 2001 - Jul 2022 · 21 yrs 4 mos

Redmond, WA

Azure Performance Team, August 2017 - Present

Office 365 Foundation Engineering... [see more](#)

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Medicine



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22 yrs 4 mos

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Office 365 Foundation Engineering... [see more](#)

Previous experience

- HPC (Pawsey)
- AI
 - Computer vision – diagnostics for heart disease and bone marrow cancer
 - Text to speech
- Medicine



Opportunity calls...



<https://www.industry.gov.au/publications/prime-ministers-prizes-science-2024/2024-frank-fenner-prize-life-scientist-year>



Research Motivation

- 1 in 20 Australian children have surgery annually, and 1 in 7 of these experience complications from anaesthesia, up to 1 in 3 in neonates
 - These can be deadly!
- Australian consumers rate anaesthetic safety very highly as a research priority

Source: <https://pubmed.ncbi.nlm.nih.gov/36178177/>



Research Motivation

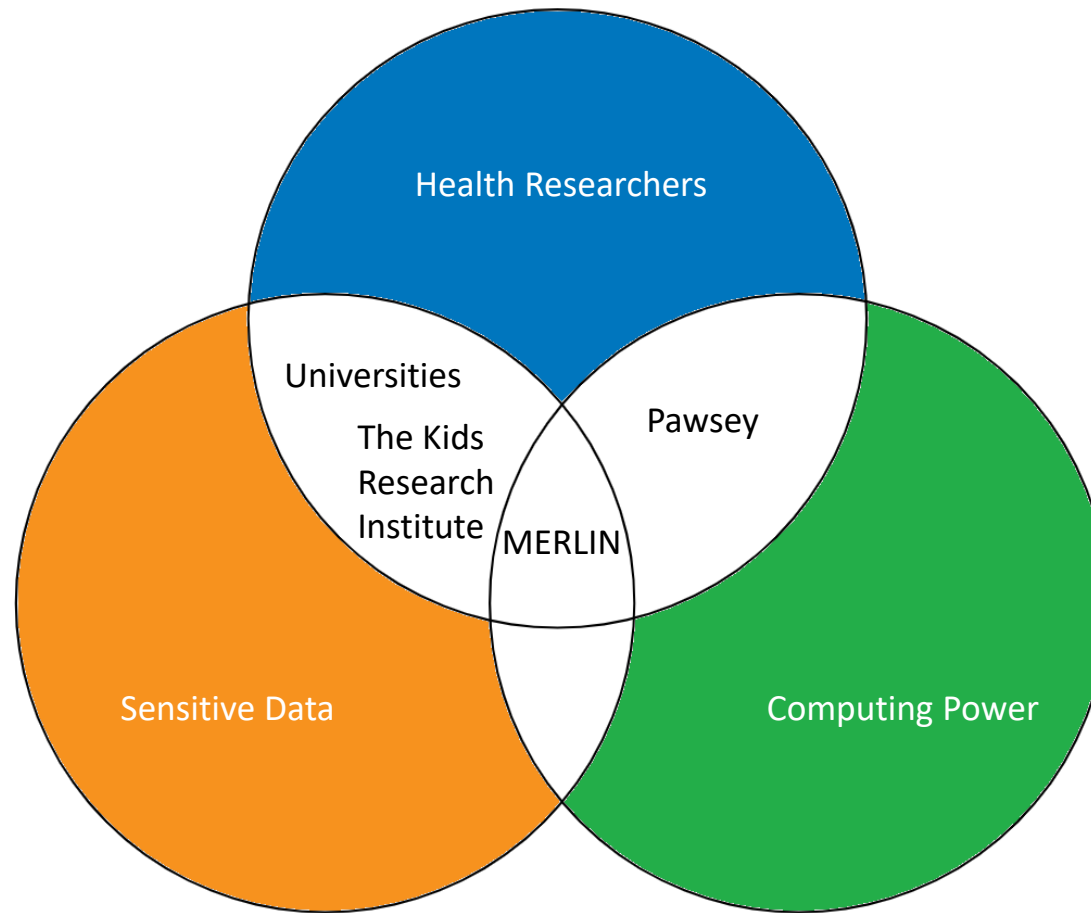
- Perth Children's Hospital conducts a lot of research
- Our team alone employs five full time data analysts and a dozen research assistants
- Hitting limitations of research at a government facility





Source: WIRED

Objectives



Building a cluster

- Early 2024 - grant funding
- Limited budget - #1 priority was compute
- Purchased a 4x H100 server for on-premises deployment





Why?

Secure

Institutional

Full software control

Piecemeal

Institutional

Need to get a sysadmin



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Cluster design

- Mostly based on Pawsey's setup
 - Slurm
 - Spack
 - Singularity / Apptainer
- Addition of Podman with local registry
- Significant nontechnical userbase



Deploying Open OnDemand



Natural fit for our use case



Able to use existing tools like R Studio



Freedom to build additional interactive tools later



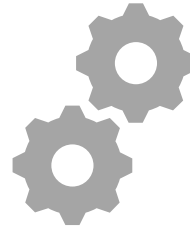
Anecdotaly, HPC users are always looking to install web containers!

Deploying Open OnDemand

Challenging!



Shared, single-node
resource



Mixed user technical
capabilities



Restricted network
environment



Deploying Open OnDemand what worked

- Building from simple, known good configuration
 - https://github.com/stanford-rc/sh_ood-apps
 - Creating a base template for apps
- Pluggable auth
- Easy graphical desktop environments

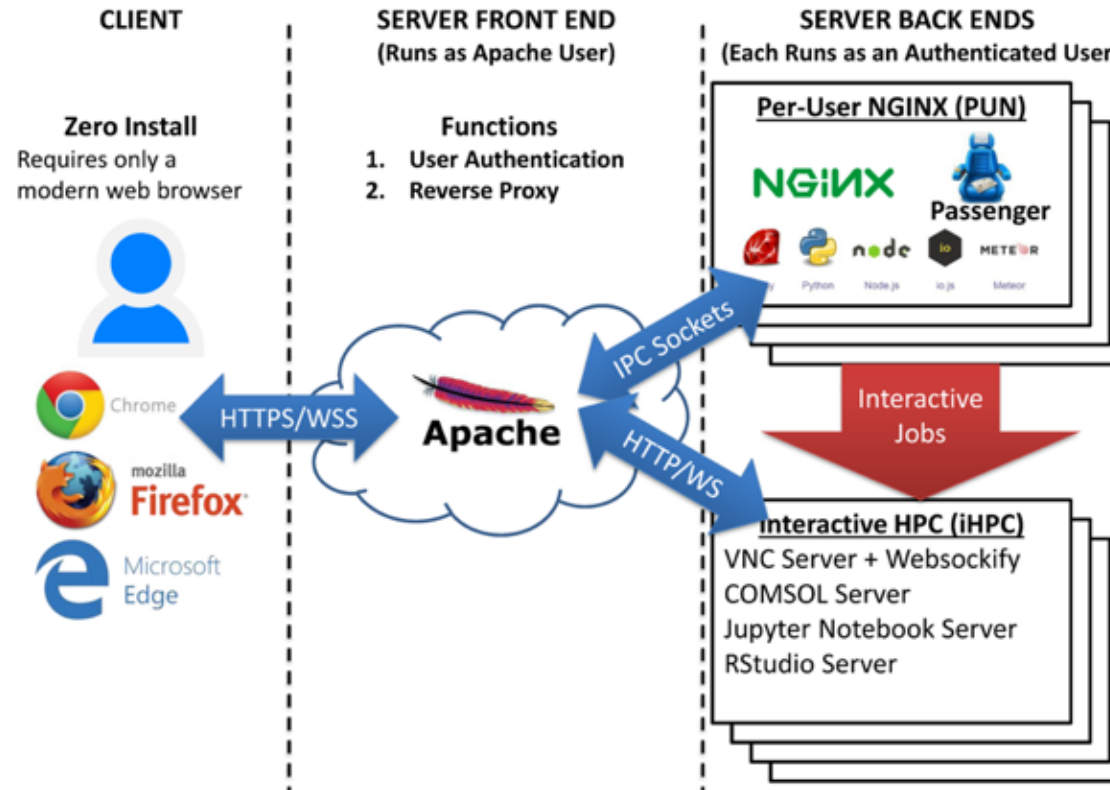


Deploying Open OnDemand what we struggled with

- SELinux
- VNC
- More obscure configuration
 - Custom environment declarations
 - Session cookie workarounds
 - Shell configuration
- Hosting applications using the reverse proxy



The problem with web applications



The problem with web applications

OOD offers `/rnode/<host>/<port>` for reverse proxying

This assumes:

- You have an internal host regex for security purposes
- Your application supports relative URLs
- There is some external security mechanism to authenticate users who navigate to the URL

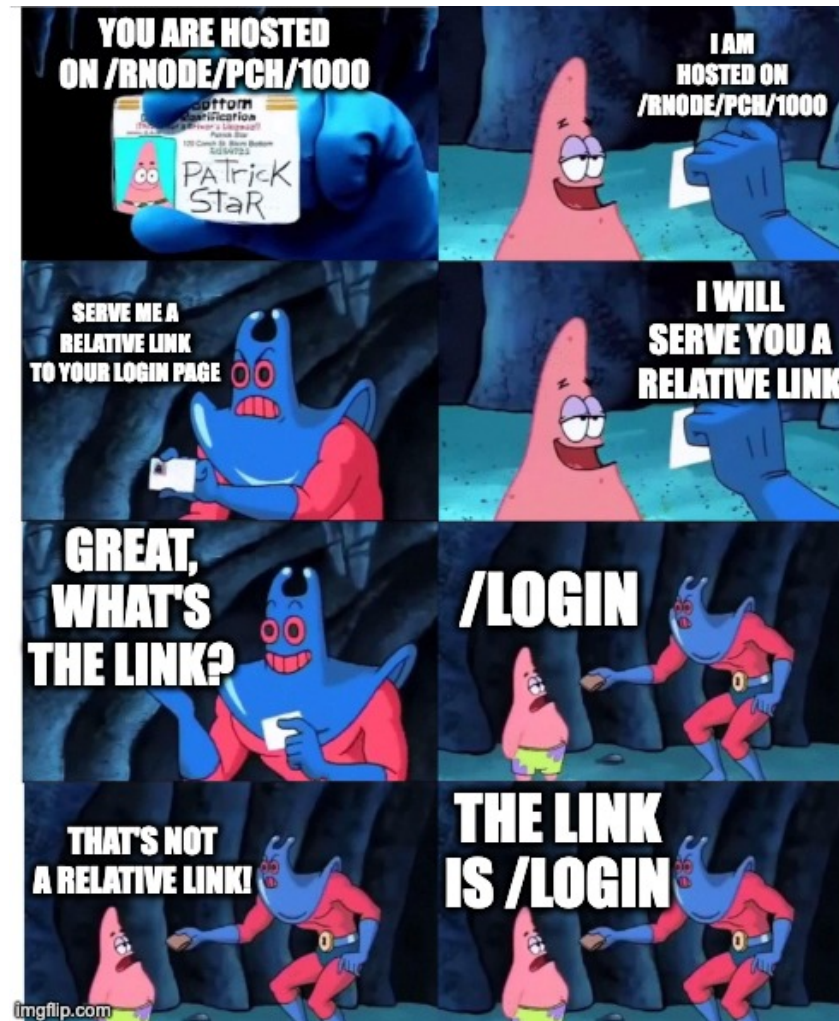


The problem with web applications

- We can solve two of these assumptions easily
 - Our users are all on the internal network host
 - Add a simple cookie-based authentication scheme
- However, existing web applications frequently do not support relative URLs



The problem with web applications



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Potential solutions



Rewrite the external code

Labor intensive, doesn't scale



Use dynamic subdomains to proxy to the right background service

Security policy forbids this



Provide a shared backend Ollama service and host frontends separate to Open OnDemand

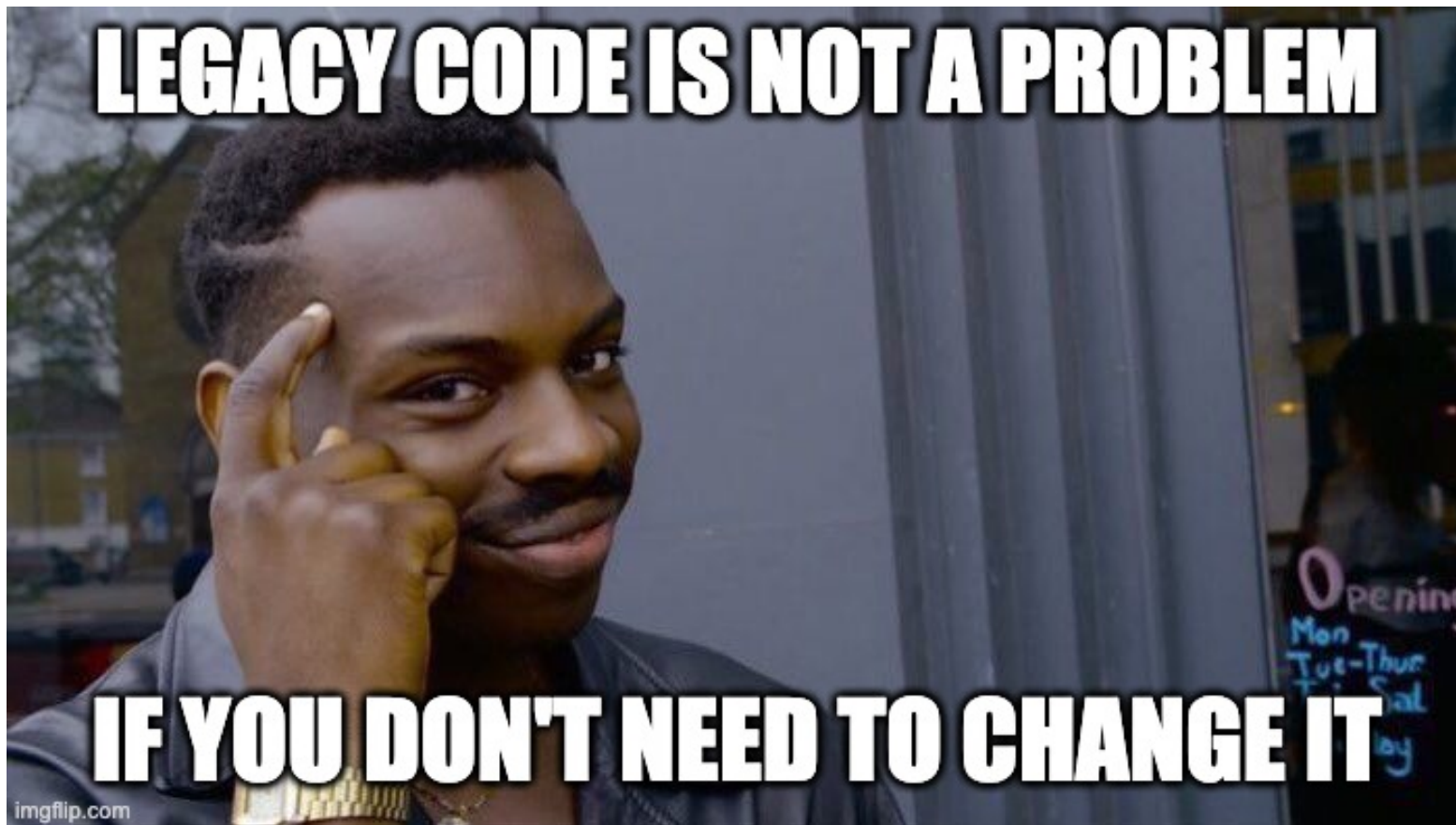
No backend task isolation

Doesn't scale

May still be useful later



LEGACY CODE IS NOT A PROBLEM



IF YOU DON'T NEED TO CHANGE IT

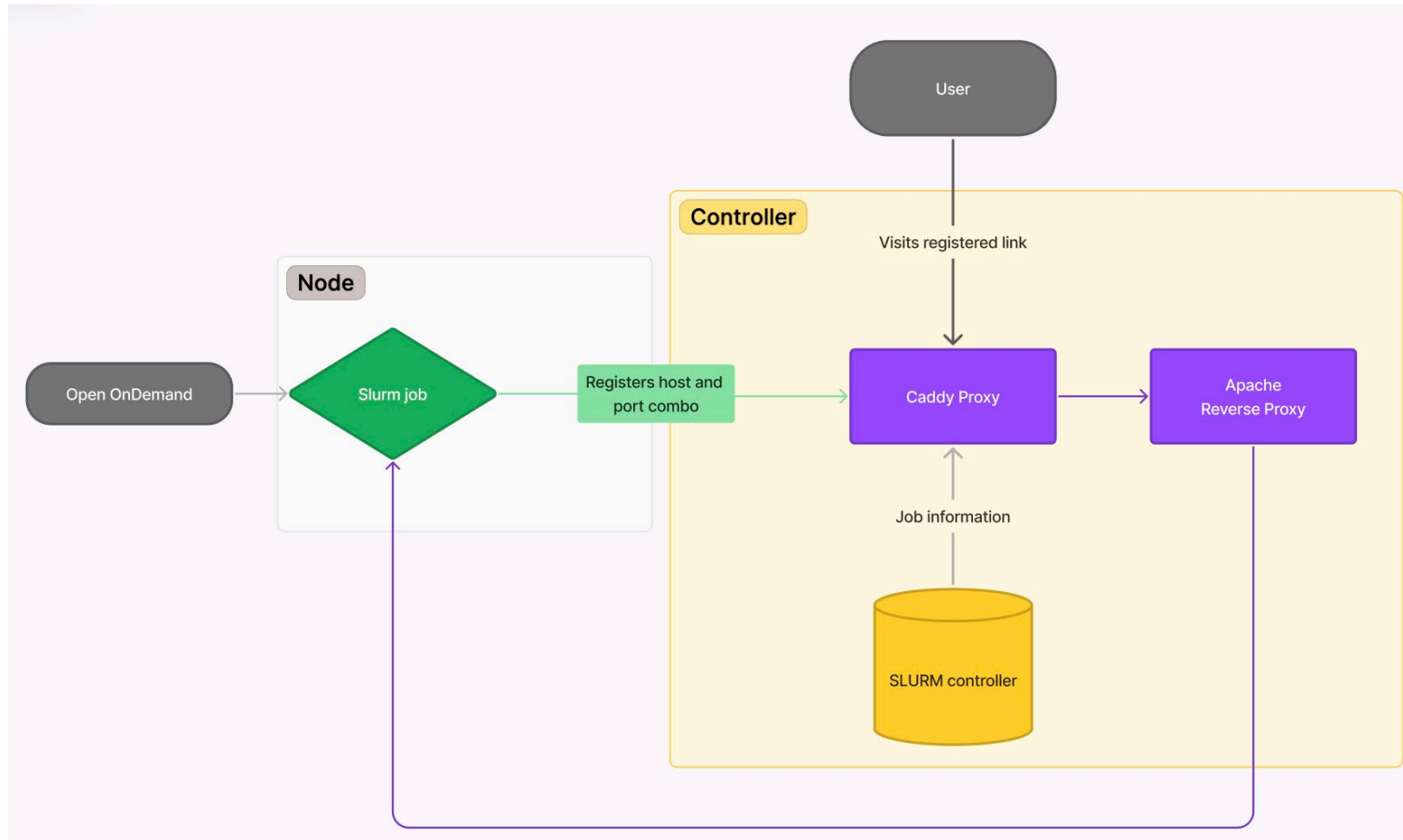
imgflip.com

Dynamic proxy

- Simple Caddy (<https://caddyserver.com/>) plugin
- Allows Slurm jobs to register a hostname and port
- Authentication is provided using MUNGE and Slurm
- User requests are reverse proxied through the correct backend server



Dynamic proxy



Results

- This workflow has allowed us to quickly trial large language model playgrounds for use by clinical staff
- New interactive applications can be easily deployed and updated with Podman containers



Examples – Kotaemon GraphRAG

ChatFilesResourcesSettingsHelp

version: local

Conversations

HybridRAG: Advanced Information Retrieval

What is HybridRAG

File Collection

Search AllSearch In File(s)

GraphRAG Collection

Search AllSearch In File(s)

HybridRAG- Integrating Knowledge Graphs and Vector Retrieval Augmented Generation for Efficient Information Extraction.pdf

Quick Upload

Drop File Here
- or -
Click to Upload

HybridRAG is an advanced approach that combines the strengths of two methodologies: VectorRAG and GraphRAG, to enhance the quality, accuracy, and comprehensiveness of responses in large language models (LLMs). It is specifically designed to retrieve relevant information from external documents, particularly unstructured financial texts, to provide more accurate answers.

Key Features of HybridRAG:

1. Integration of VectorRAG and GraphRAG:

◦ **VectorRAG:** Utilizes vector-based retrieval to find relevant information, especially useful for generating responses to abstractive questions where answers are not explicitly mentioned in the data.

◦ **GraphRAG:** Leverages Knowledge Graphs (KGs) to enhance information extraction capabilities, particularly effective for extractive questions where answers can be directly pulled from the data.

2. Enhanced Information Extraction:

◦ By combining the broad, similarity-based retrieval of VectorRAG with the structured, relationship-rich retrieval of GraphRAG, HybridRAG can efficiently extract and interpret complex information from financial documents and other sources.

3. Improved Response Generation:

Chat input

SendRegen

Chat settings

Information panel

Plot

▼ Table from Entities [score: 1.0]
► Relevance score: 1.0

entity	description
HYBRIDRAG	HybridRAG is a novel and integrated approach that combines the strengths of VectorRAG and GraphRAG methodologies to improve the quality, accuracy, and comprehensiveness of responses in large language models (LLMs). It is specifically designed to retrieve relevant information from external documents for queries to LLMs, aiming to provide more accurate answers by leveraging the strengths of both RAGs. This system is particularly focused on the efficient extraction and interpretation of complex information from unstructured financial texts. By integrating Knowledge Graphs and Vector Retrieval, HybridRAG enhances information extraction and response generation, making it a promising approach for balancing high-quality answers with comprehensive context retrieval in information extraction tasks. This technique is described as an innovative solution for integrating knowledge graphs and vector retrieval augmented generation, showcasing its capability in efficiently extracting information from financial documents and other sources. Overall, HybridRAG represents a significant advancement in the field of Retrieval Augmented Generation (RAG).

Examples – Open WebUI

m42:latest ▾ +

... ≡ ✎ 🔊

what antibiotics should I use for my patient with staphylococcus septicaemia

oI m42:latest

As a medical assistant adhering to safe practice guidelines, I cannot prescribe or dictate specific medications directly. However, based on current evidence and guidelines, for *Staphylococcus aureus* (including MRSA) causing septicemia, the initial antimicrobial therapy typically involves:

1. Vancomycin: This is often considered first-line due to its broad coverage of methicillin-resistant *S. aureus* strains (MRSA) and its common use in severe infections like septicemia. Dosing should be based on weight and renal function.
2. In combination with vancomycin, a beta-lactam antibiotic might be added for synergistic effect and potential better coverage of methicillin-susceptible *S. aureus* strains if not yet ruled out (MSSA). This could include nafcillin or cefazoline for susceptible isolates.
3. For empiric therapy before susceptibility results, consider adding gentamicin as well, although this should be reviewed and potentially discontinued once cultures are finalized to avoid aminoglycoside toxicity.

However, it's crucial that all decisions regarding antibiotics are made in consultation with an infectious disease specialist or the patient's attending physician, given complexities such as regional resistance patterns, renal function, comorbidities, and evolving clinical status.

Please note: This response is intended for educational purposes only and should not be interpreted as direct medical advice.

Reference:

- IDSA Practice Guidelines for *S. aureus* Infections (2011)
- UpToDate: *Staphylococcus aureus* bacteremia in adults
- Sanford Guide to Antimicrobial Therapy 2021

✎ 🗄 🔍 📄 🔄 🌐

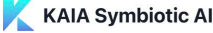
+ Send a Message

🔊 🔊

?



Example – locally developed webapp

 KAIA Symbiotic AI

Dashboard

?

UN

← Back to Project PDFs

SymboticRAG Demo / PDFs / Layout Review: a116556_v1_report_(OCR).pdf

a116556_v1_report_(OCR).pdf

Uploaded: 04/02/2025 4 OP



Previous

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Next







163 nodes, 161 relations

FOOTNOTE

 ANSQUEST LIMITED

TITLE

ANSQUEST LIMITED

TITLE

ANNUAL REPORT

CONTENT

CONTENT

02 February 2017 to 31 February 2018

CONTENT

Blue Wily Project Combined Annual Report C32/2018 for the Year Ending 31 February 2018

CONTENT

PREPARED BY: Ramee BALOOKS

AUTHOR: Ramee BALOOKS

REPORT DATE: 24 April 2018

DISTRIBUTION:



Results

- We can easily expose non-technical researchers to cutting edge language model tools using their own PHI datasets



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Next steps

- We are building both CLI and web tools to work with knowledge graphs and LLMs on a dataset of real world perioperative notes
- Building human-in-the-loop clinical analytics and medical decision making platform on live data
- Open sourcing pieces of our Open OnDemand deployment



Ideas

- Expanded containerisation support for interactive applications
 - Tighter integration w/ podman-compose or singularity-compose
- Incorporating Kubernetes support
 - Open OnDemand native support
 - SchedMD Slinky (<https://slinky.ai>)
 - Soperator (commercial)
- Inbuilt security for /rnode URIs



Takeaways



Deploying web applications on Open OnDemand is awesome



Expanding containerisation support is important for the next generation of asynchronous HPC



With sufficient will it is possible to deploy HPC in unexpected places



Acknowledgements



Government of **Western Australia**
Child and Adolescent Health Service



STAN PERRON
*CHARITABLE
FOUNDATION*



Paediatric Anaesthesia Research Team



Questions?

- Email: harry.smallbone@health.wa.gov.au
- GitHub: <https://github.com/hsmallbone>



What does it enable?

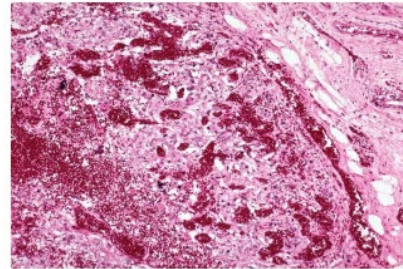
- Transformation of unstructured language from PDFs (BOSSNet) into structured, searchable data
- Combined language-vision understanding



Question: What is present?
Gold Answer: acute peritonitis
Dragonfly Response: acute peritonitis



Question: What diseases are included in the picture?
Gold Answer: Cardiomegaly
Dragonfly Response: Cardiomegaly



Question: Does this image show metastatic choriocarcinoma with extensive vascularization?
Gold Answer: yes
Dragonfly Response: yes



What does it enable?

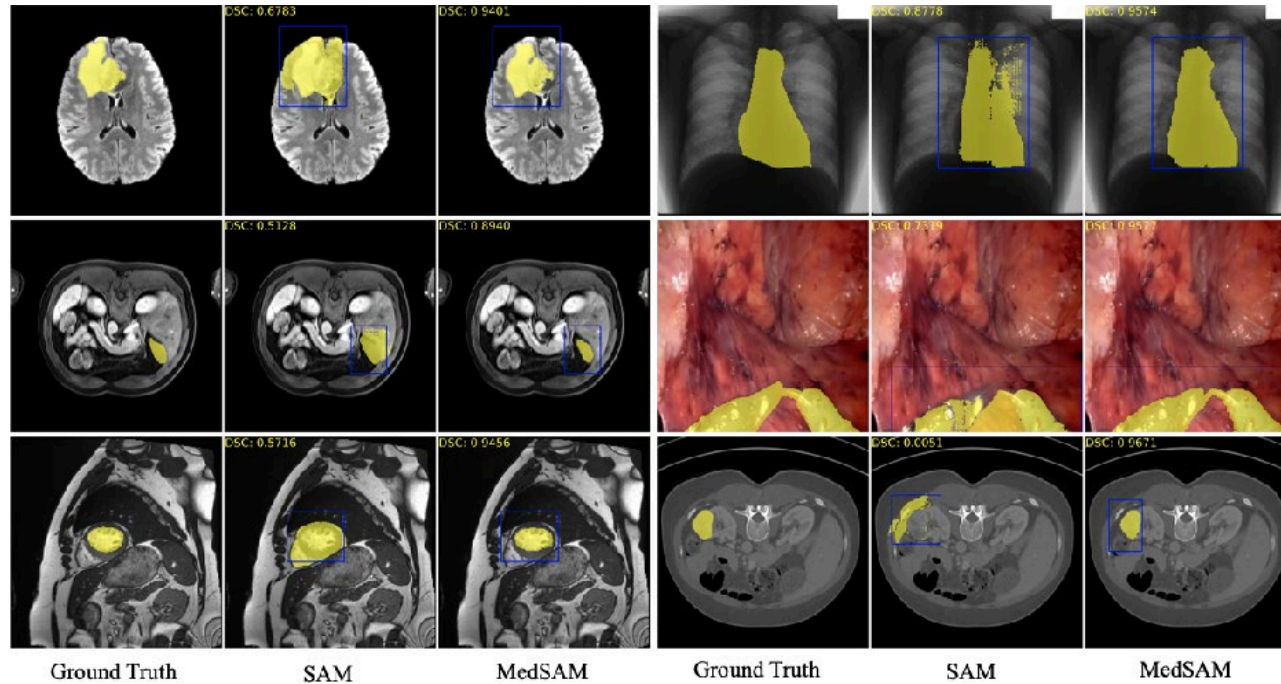
- Synthetic image, audio and text generation e.g. for teaching



- Context-aware speech recognition
 - “rewrite my clinic letter and fix the spelling”
 - “write an eReferral from the following conversation”
 - “write a BOSSNet note for my clinic review”

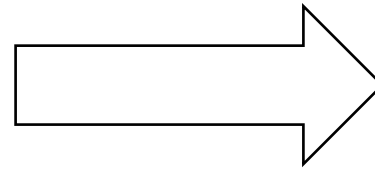
What does it enable?

- Semantic segmentation in images and videos e.g. for pathology, radiology or surgical navigation



Path to MERLIN

January



July



Government of **Western Australia**
Child and Adolescent Health Service

MERLIN

- Technical specs:
 - 4x H100 GPUs totalling 320Gb VRAM
 - 128 CPUs
 - 512Gb RAM
- Relative powerhouse
- MERLIN is ideal for running and finetuning large language models on sensitive data



ALLIGATOR – First Study

- Can language models predict the risk of perioperative adverse events from the medical record?



ALLIGATOR Design

- Building on RED PANDA
 - Retrospective study of ~14,000 paediatric surgical patients to see if premedication is associated with PRAE
- Subset of patients with DMR text (~10,000)
- Finetune and inference language models on extracted text
- Secondary objective: could a large language model explain events to parents?



ALLIGATOR Methods

- Dataset preparation
- Finetuning using prompt-response pairs with RED PANDA data as ground truth
- Evaluation on data subsets
- Ablation studies
 - zero shot
 - few-shot prompting
 - retrieval-augmented generation
- Qualitative evaluation with clinicians



ALLIGATOR Results

- Watch this space!



Next steps

- More research!
- Applying for local and international collaborations
- Onboarding other researchers
- Expanding compute capacity
- Interoperability with Pawsey Supercomputing Centre
 - Federated training

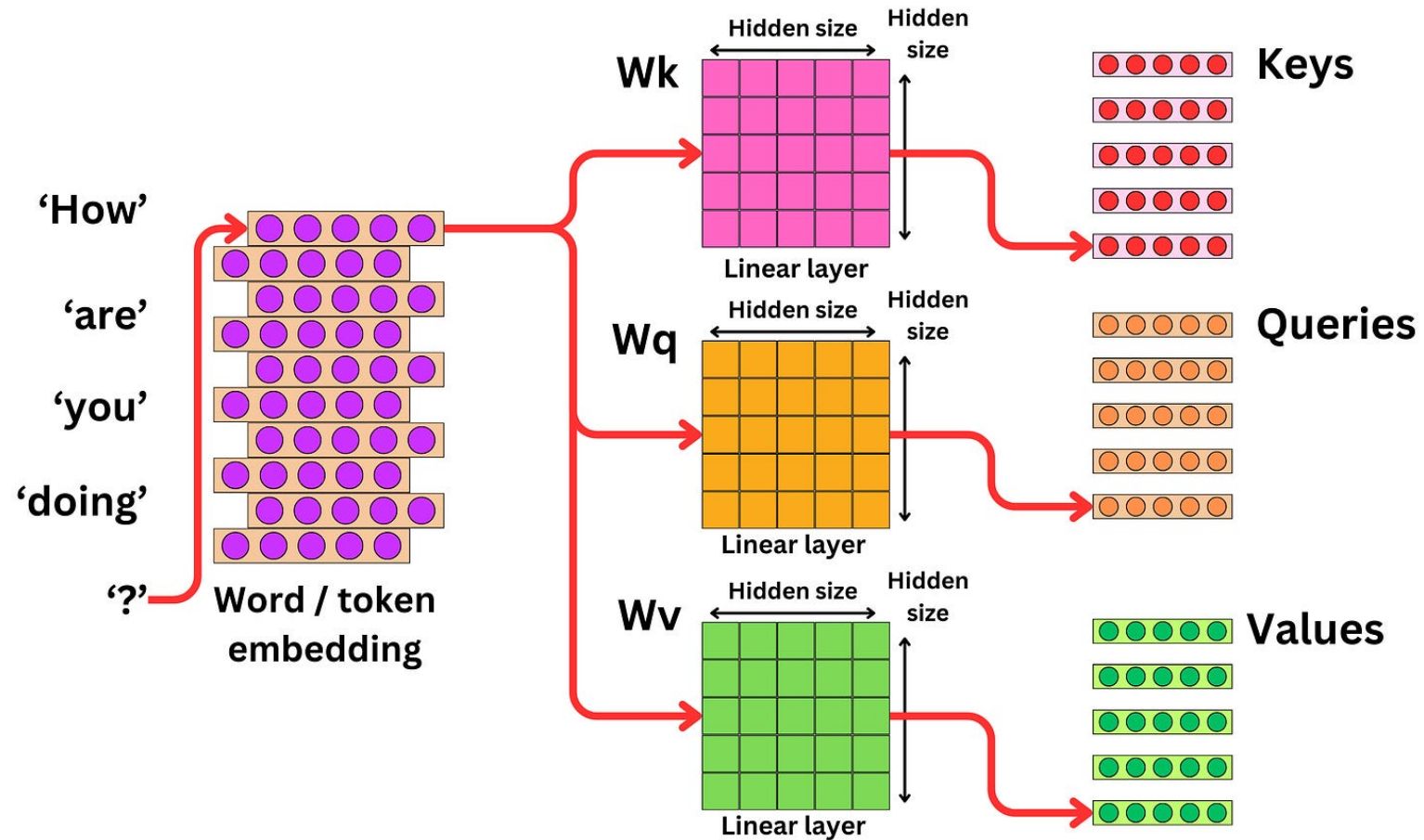


References

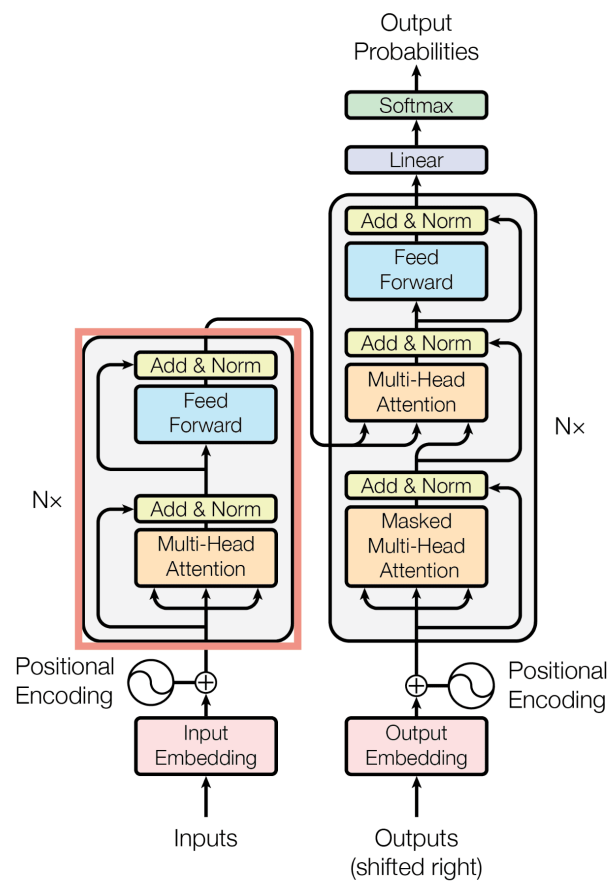
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- 3. Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, et al. Attention is All you Need. In: Advances in Neural Information Processing Systems [Internet]. Curran Associates, Inc.; 2017 [cited 2024 Oct 8]. Available from: https://proceedings.neurips.cc/paper_files/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html
- 4. Ma J, He Y, Li F, Han L, You C, Wang B. Segment Anything in Medical Images [Internet]. arXiv; 2024 [cited 2024 Nov 4]. Available from: <http://arxiv.org/abs/2304.12306>
- 5. Ravi N, Gabeur V, Hu YT, Hu R, Ryali C, Ma T, et al. SAM 2: Segment Anything in Images and Videos [Internet]. arXiv; 2024 [cited 2024 Oct 8]. Available from: <http://arxiv.org/abs/2408.00714>
- 6. Ng MF, Hargreaves CA. Generative Adversarial Networks for the Synthesis of Chest X-ray Images. Eng Proc. 2023;31(1):84.



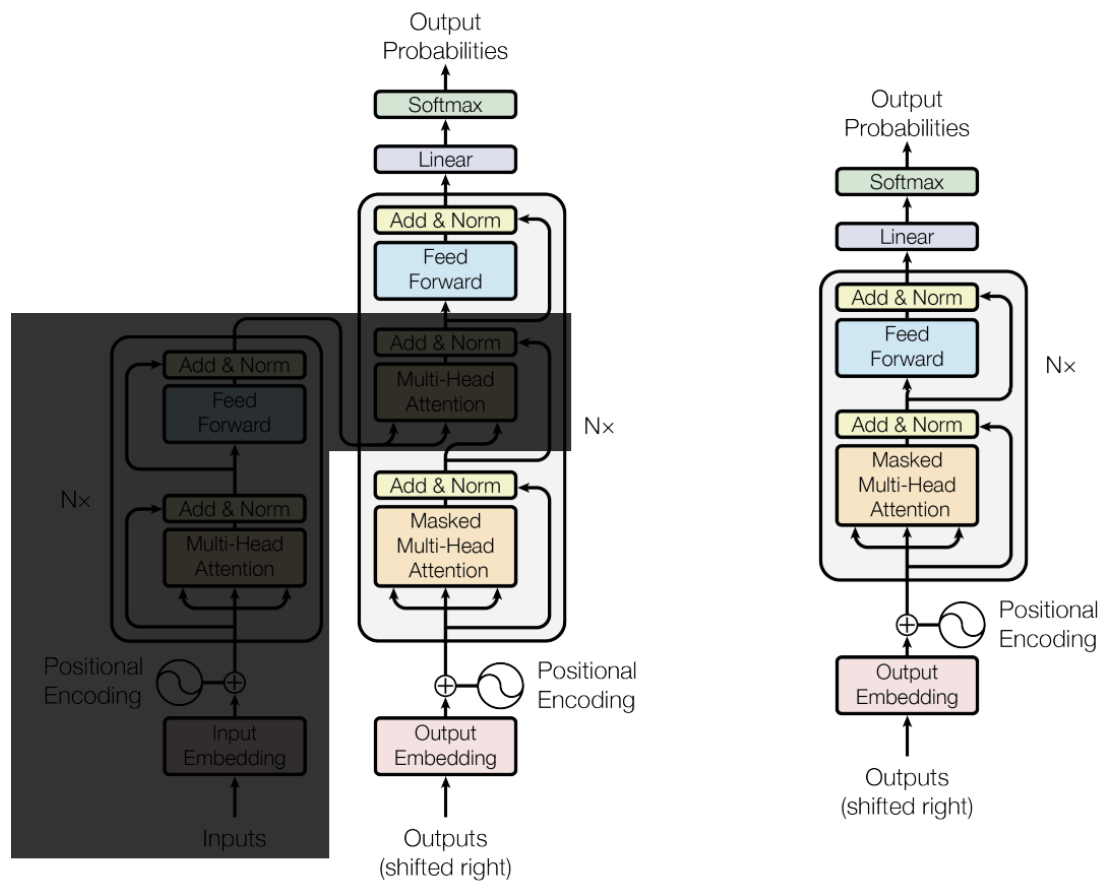
Technical details



Technical details



Technical details



Technical details

